Open-Sora 2.0: Training a Commercial-Level Video Generation Model in $200k

Open-Sora Team

HPC-AI Tech

Abstract

Video generation models have achieved remarkable progress in the past year. The quality of AI video continues to improve, but at the cost of larger model size, increased data quantity, and greater demand for training compute. In this report, we present Open-Sora 2.0, a commercial-level video generation model trained for only $200k. With this model, we demonstrate that the cost of training a top-performing video generation model is highly controllable. We detail all techniques that contribute to this efficiency breakthrough, including data curation, model architecture, training strategy, and system optimization. According to human evaluation results and VBench scores, Open-Sora 2.0 is comparable to global leading video generation models including the open-source HunyuanVideo and the closed-source Runway Gen-3 Alpha. By making Open-Sora 2.0 fully open-source, we aim to democratize access to advanced video generation technology, fostering broader innovation and creativity in content creation. All resources are publicly available at: https://github.com/hpcaitech/Open-Sora.

1 Introduction

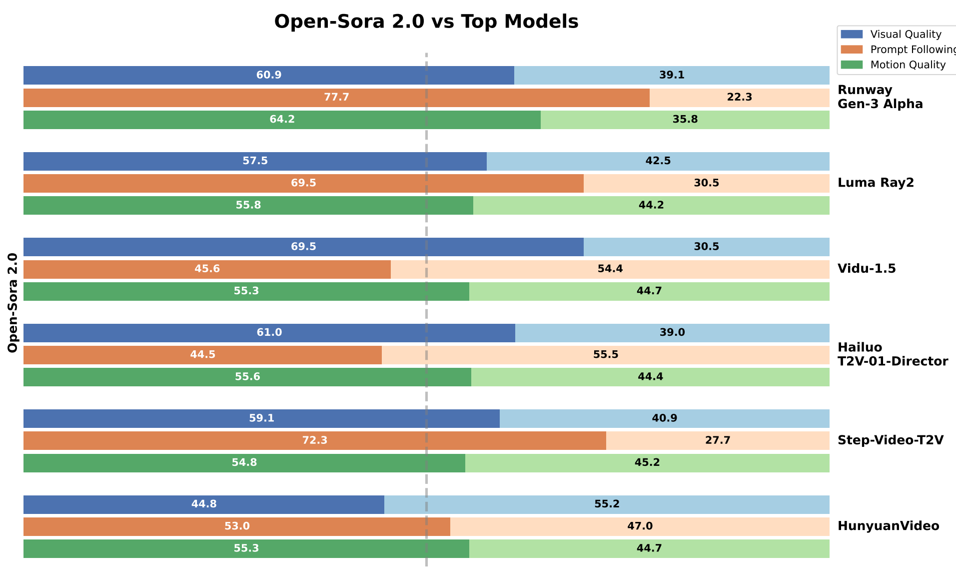

Figure 1: Human preference evaluation of Open-Sora 2.0 against other leading video generation models. Win rate represents the percentage of comparisons where our model was preferred over the competing model. The evaluation is conducted on 100 prompts carefully designed to cover three key aspects: 1) visual quality, 2) prompt adherence, and 3) motion quality. Results show that our model performs favorably against other top-performing models in all three aspects.

The past year has witnessed an explosion of video generation models. Since the emergence of OpenAI's Sora [4] in February 2024, a series of video generation models—either open-source [16, 19, 22, 27, 44] or proprietary [26, 32, 35]—have appeared at an unprecedented pace, each striving to achieve "Sora-level" generation quality. While the quality of generated videos continues to improve, there is a clear trend toward rapid growth in model size, data quantity, and computing resources. Following the success of scaling large language models (LLMs) [15, 18], researchers are now applying similar scaling principles [19] to video generation, converging on similar model architectures and training techniques.

In this report, however, we show that a top-performing video generation model can be trained at a highly controlled cost, offering new insights into cost-effective training and efficiency optimization.. We build Open-Sora 2.0, a commercial-level video generation model trained for only $200k. As shown in Table. 7, the training cost is 5-10 times lower than comparable models like MovieGen [32] and Step-Video-T2V [27] based on a fair comparison. This remarkable cost efficiency stems from our joint optimization of data curation, training strategy, and AI infrastructure—all of which are detailed in the following sections.

We show the human preference evaluation of Open-Sora 2.0 and other global leading video generation models in Figure 1. The models for comparison include proprietary models like Runway Gen-3 Alpha [35], Luma Ray2 [26] and open-source models like HunyuanVideo [19] and Step-Video-T2V [27]. We evaluate these models in three important aspects: 1) visual quality, 2) prompt adherence, and 3) motion quality. Despite the low cost of our model, it outperforms these top-performing models in at least two aspects out of the three, demonstrating its strong potential for commercial deployment.

This report is structured as follows. In Section 2, we elaborate on our data strategy, including our hierarchical data filtering system and annotation methods. Section 3 details our model architecture, covering both our novel Video DC-AE autoencoder design and the DiT architecture. Section 4 explores our cost-effective training methodology, which enables commercial-level quality at just $200k. Section 5 presents our conditioning approaches, including image-to-video and motion control techniques. Section 6 outlines the system optimizations that maximize training efficiency, and Section 7 evaluates our model's performance against leading video generation models. Finally, we conclude with a summary of our contributions and insights for future research.

2 Data

Our goal for data is to build a hierarchical data pyramid, catering to the requirement of the progressive training process. To this end, we develop a collection of filters that function distinctly from each other, aiming to tackle various types of data detections. By progressively strengthening the degree of filtration, we can obtain subsets of smaller sizes but higher purity and quality.

In Section 2.1, we first elaborate the data filtering system that consists of two main components: preprocessing and score filtering. Then, we introduce data annotation methods in Section 2.2. Finally, we present detailed statistics of the whole dataset in Section 2.3.

2.1 Data Filtering

The hierarchical data filtering system is illustrated in Figure 2. We begin by preprocessing raw videos into trainable video clips, then progressively apply a series of filters, ranging from loose to strict, to construct a structured data pyramid.

2.1.1 Preprocessing

The preprocessing phase converts raw videos into short clips suitable for training. During this phase, we first eliminate broken files and raw videos with outlying attributes. Specifically, we filter out videos with a duration of less than 2 seconds, a bit per pixel (bpp) below 0.02, a frame rate (fps) under 16, and an aspect ratio outside the range [1/3, 3], and those with a "Constrained Baseline" profile.

Then, we detect continuous shots within the raw video and segment it into short clips based on the detection result. For shot detection, we apply the libavfilter from FFmpeg [12] to calculate the scene scores, which quantify visual differences among frames. The shot boundaries are identified at points where the change in scene scores exceeds an empirically determined threshold.

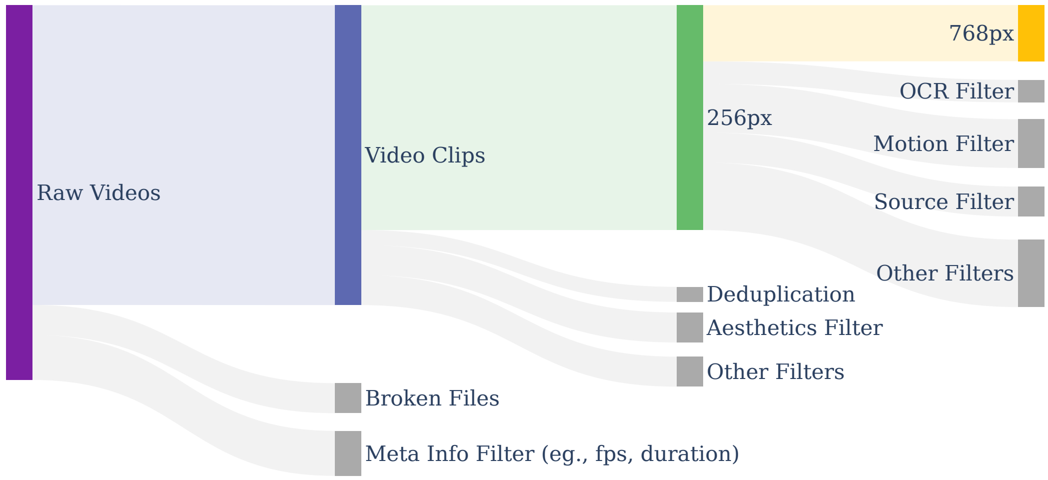

Figure 2: **The hierarchical data filtering pipeline**. The raw videos are first transformed into trainable video clips. Then, we apply various complimentary score filters to obtain data subsets for each training stage.

Finally, we segment the video based on the shot detection results, ensuring that the output clips adhere to specific format constraints: frame rate (fps) below 30, longer dimension not exceeding 1080 pixels, and H.264 codec. Additionally, black borders are removed during this process. For those shots exceeding 8 seconds, we further divide them into multiple 8-second clips, while shots shorter than 2 seconds are discarded.

2.1.2 Score Filtering

To address the various defects in raw data, we develop a bag of complimentary filters, each targeting a specific aspect of data quality. These filters work together as a comprehensive and robust purification system. Typically, each filter evaluates a sample by assigning a score based on its respective criteria, and the filtering intensity is controlled by a threshold. We introduce all the score-based filters in the following sections.

Aesthetic Score. We assess the aesthetic quality of a sample using the CLIP+MLP aesthetic score predictor [36]. For a video sample, we extract the first, middle, and last frame, compute their individual scores, and take the average as the final aesthetic score.

Motion Score. The motion intensity of a video is measured using the VMAF motion score from FFmpeg's libavfilter library [12]. Videos with extremely low or excessively high motion scores are filtered out to maintain a balanced motion range.

Blur Detection. To evaluate image clarity, we apply the Laplacian operator from OpenCV [3]. A sample is considered blurry if the variance of the Laplacian image falls below a configurable threshold. For video samples, we extract five uniformly spaced frames and adopt a majority voting approach to determine blurriness.

OCR. We detect text bounding boxes using PaddleOCR [2] and compute the total area of bounding boxes with a confidence score above 0.7. Images or videos containing excessive text are discarded to avoid unwanted overlays.

Camera Jitter Detection. We detect camera jitter using Shot Boundary Detection from the PySceneDetect library [6]. A video is identified as having camera jitter if the average frame-to-frame change exceeds a predefined threshold.

대한민국

대한민국

국장

국가: 애국가

0:00 / 0:00

.php?language=ko&pagename=%EB%8C%80%ED%95%9C%EB%AF%BC%EA%B5%AD&params=37.568889_N_126.976667_E_)

대한민국(대한민국)

정치

역사

지리

인문

인구

경제

종교
기타

대한민국(한국 한자: 大韓民國)은 동아시아의 한반도 군사 분계선 남부에 위치한 나라이다. 약칭으로 **한국**(한국 한자: 韓國), 별칭으로 **남한**(한국 한자: 南韓, 문화어: 남조선)이라 부르며 현정체제는 대한민국 제6공화국이다. 대한민국의 국기는 대한민국 국기법에 따라 태극기이며[6], 국가는 관습상 애국가, 국화는 관습상 무궁화이다. 공용어는 한국어이며 수도는 서울특별시이다. 인구는 2024년 2월 기준으로 5,130만명이고[7], 이 중 절반이 넘는(50.74%) 2,603만명이 수도권에 거주한다.[8]

대한민국은 1948년 5월 10일 총선거를 통해 제헌국회를 구성하였고, 1948년 8월 15일 대한민국 정부를 수립하였다. 대한민국은 20세기 후반 이후 급격한 경제 성장을 이루었으나, 그 과정에서 1990년대말 외환 위기 등의 부침이 있기도 했다. 현재는, 2024년 국내총생산(GDP) 기준 세계 12위권의 경제 규모를 가지고 있으며, 2024년 1인당 국민 총소득(GNI)은 명목 3만 6,624달러이다.[9] 2023년 유엔개발계획(UNDP)이 매년 발표하는 인간개발지수 (HDI) 조사에서 세계 20위를 기록하였다.[3] 2021년 7월 2일 스위스 제네바 본부에서 열린 '제68차 무역개발이사회' 마지막 회의에서 한국의 지위를 선진국 그룹으로 '의견 일치'로 변경하고 선진국으로 인정했다.[10] 다만 높은 자살률, 장시간 근로 문화와 높은 산업 재해 사망률, 저출산 등의 사회 문제가 이 같은 성과와 병존한다. 민주주의 수준의 경우, 이코노미스트에서 발표하는 민주주의 지수 조사에서 2019년 기준 23위의 8.0점을 기록한 바와 같이 아시아에서 그 수준이 가장 높은 국가 가운데 하나이다. 대한민국은 주요 20개국(G20), 경제 협력 개발 기구(OECD), 개발 원조 위원회(DAC), 파리 클럽 등의 회원국으로 활동하고 있다.

1948년 이후로 오늘날까지 한반도에는 대한민국과 조선민주주의인민공화국이라는 두 개의 분단국가가 각각 남북에 위치한다. 한반도와 부속도서의 면적은 약 22만 km^2이며, 인구는 대한민국과 조선민주주의인민공화국을 합쳐 2019년을 기준으로 약 7,718만 명에 달한다.[11]

국호

어원

대한민국(大韓民國) 국호 중 '**한**(韓)'의 어원은 본래 국가의 명칭이 아니라, 군주를 뜻하는 칭호인 '**한**(汗)'에서 비롯되었다.